Noise-adapted Quantum Error Correction
for Non-Markovian Noise

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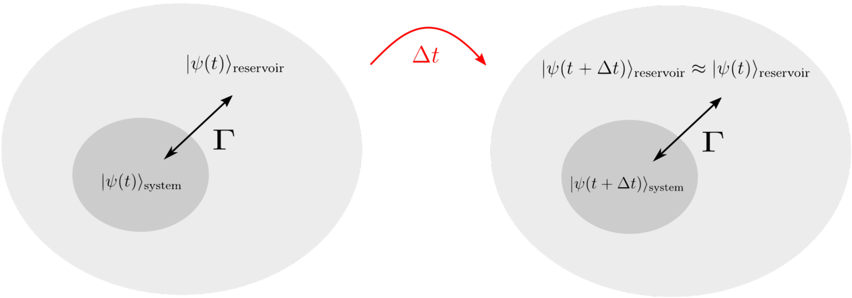
In this talk

We will:

- begin by describing non-Markovianity and based on a set of assumptions, arrive at a theoretical framework.
- understand why non-Markovianity is a big issue when trying to achieve efficient error correction.
- look at non-Markovian ways of error correction, notably the **Petz** recovery channel, and its application on two simple models: Amplitude Damping (AD) and a Dephasing channel.

Outline

Primer: Open Quantum Systems



- Quantum systems don't exist in isolation, they interact with their environment, the *bath*.

Primer: Open Quantum Systems (Contd.)

- Under a series of assumptions, their evolution can be described by the Gorini–Kossakowski–Sudarshan–Lindblad Master Equation

$$\dot{\rho} = \mathcal{L}[\rho] := -\frac{i}{\hbar}[H, \rho] + \sum_i \gamma_i \left(L_i \rho L_i^\dagger - \tfrac{1}{2}\{L_j^\dagger L_j, \rho\} \right), \quad \gamma_i \geq 0 \tag{1}$$

- We can also write their evolution as the (already familiar to us) completely positive, trace-preserving (CPTP) maps.

$$\sigma = \mathcal{E}\left[\rho_S\right] := \mathrm{tr}_E \left[U_{SE} \left(\rho_S \otimes \rho_E \right) U_{SE}^\dagger \right] \tag{2}$$

$$= \sum_i K_i \rho K_i^\dagger, \tag{3}$$

where $\{K_i\}$ satisfy $\sum_i K_i^\dagger K_i = I$ and are called the **Kraus-Sudarshan** (KS) operators.

Primer: Markovian Dynamics

- The master equation dynamics assume:
  - **Factorizability** of initial state.
  - The **Born-Markov approximation**: system timescale $\gg$ environment timescale.
  - **Rotating wave approximation** (RWA): fast rotating terms in the Hamiltonian are ignored.
- Fact: if we relax (i) or (iii), the system is not guaranteed to be a CP map. Upon relaxing (ii), however, we can allow for time-dependence:

$$\dot{\rho} = \mathcal{L}(t)[\rho] := -\frac{i}{\hbar}[H, \rho] + \sum_i \gamma_i(t) \left(L_i(t) \rho L_i^\dagger(t) - \tfrac{1}{2}\{L_j^\dagger(t) L_j(t), \rho\} \right) \tag{4}$$

- Moreover, $\gamma_i(t)$ need not be $\geq 0 \ \forall \ t$.
- Markovianity arises as a **special case** of the above, when L_i are constant, and γ_i are constant and non-negative.

Primer: Non-Markovian Dynamics

- In the Markovian case, we denote $\mathcal{E}_M(t) = \exp(t\mathcal{L})$, and the system state evolves as $\rho(t) = \mathcal{E}_M(t)[\rho(0)]$. $\mathcal{E}_M(t)$ follows a one-parameter semi-group structure:

$$\mathcal{E}_M(t_2 + t_1) = \mathcal{E}_M(t_2)\mathcal{E}_M(t_1) \tag{5}$$

- If the decay rates γ_i are time-dependent, the map becomes

$$\mathcal{E}(t, t_0) = \mathcal{T} \exp\left(\int_{t_0}^{t} \mathcal{L}(t)dt\right), \quad \mathcal{T} : \text{time-ordering} \tag{6}$$

  If any one of the decay rates $\gamma_i(t)$ becomes less than 0 for any interval of time, then the dynamics become strictly Non-Markovian.

- For that interval of time, we see interesting effects such as **information backflow**, a type of gain of coherence in the system.

Non-Markovian Dynamics: Example

Assume the damped **Jaynes-Cummings**(JC) model of a two-level system (TLS) interacting with a bosonic reservoir at 0K. Here,

$$H_{tot} = \frac{\omega_0 \sigma_z}{2} + \sum_j \omega_j a_j^\dagger a_j + \sum_j (g_j \sigma_+ a_j + g_j^* \sigma_- a_j^\dagger), \tag{7}$$

and the bath spectral density

$$J(\omega) = \frac{\Gamma_0 b^2}{2\pi[(\omega_0 - \Delta - \omega)^2 + b^2]}, \quad \Delta = \omega_0 - \omega_c, \tag{8}$$

where ω_0 & ω_c are the TLS frequency and the bath central frequency respectively, and b the spectral bandwidth.

Non-Markovian Dynamics: Example (Contd.)

Applying

$$\frac{d}{dt}\rho_{SE} = -\frac{i}{\hbar}[H_{tot}, \rho_{SE}],$$

and taking the partial trace, we get

$$E_1(t) = \begin{pmatrix} 1 & 0 \\ 0 & \sqrt{1-\gamma(t)} \end{pmatrix} \quad ; \quad E_2(t) = \begin{pmatrix} 0 & \sqrt{\gamma(t)} \\ 0 & 0 \end{pmatrix}$$

with damping parameter:

$$\gamma(t) = 1 - |G(t)|^2, \quad G(t) = e^{-bt/2}\left(\tfrac{b}{d}\sinh\left(\tfrac{dt}{2}\right) + \cosh\left(\tfrac{dt}{2}\right)\right), \quad d = \sqrt{b^2 - 2\Gamma_0 b}$$

The system undergoes non-Markovian evolution when $b \ll 2\Gamma_0$, when $b \gg 2\Gamma_0$ the dynamics turn Markovian and it has a Lindblad form with a constant decay rate:

$$\Gamma(t) = \Gamma_0$$

We shall see more of this system later.

Sidebar: CP-divisibility

- A map is **Completely Positive(CP)** iff its *Choi-Jamiołkowski* matrix is positive semi-definite, i.e. $\chi = [\mathcal{E} \otimes I][\rho] \geq 0$, ρ being maximally entangled. CP maps can be written in the SK form discussed earlier:

$$\rho(t) = \mathcal{E}(t)[\rho(0)] = \sum_i E_i(t)\rho(0)E_i^\dagger(t)$$

- **CP-divisibility** is achieved when a map $\mathcal{E}(t_2, t_0)$ can be written as $\mathcal{E}(t_2, t_1)\mathcal{E}(t_1, t_0)$, both CP, for all $t_2 \geq t_1 \geq t_0$.
- Markovian channels $\mathcal{E}_M(t)$ are always **CP-divisible**.
- Non-Markovian channels $\mathcal{E}_{NM}(t, t_0)$ may not be **CP-divisible**, i.e. if we write $\mathcal{E}_{NM}(t_2, t_0) = \mathcal{E}_{NM}(t_2, t_1)\mathcal{E}_{NM}(t_1, t_0)$. $\mathcal{E}_{NM}(t_2, t_1)$ is not always completely positive for all t_1. However, since $\mathrm{tr}_E\left[\left(U_{SE}\left(\rho_S \otimes \rho_E\right)U_{SE}^\dagger\right)^\dagger\right] = \mathrm{tr}_E\left[U_{SE}\left(\rho_S \otimes \rho_E\right)U_{SE}^\dagger\right]$, it falls under the category of Hermiticity-preserving, trace preserving (HPTP) maps.

Issue with Non-Markovian Dynamics

- Standard QEC assumes:
  - Uncorrelated (independent) errors
  - Local CPTP noise acting in time steps
  - Irreversible information loss (monotonic decoherence)
- However,
  - **Terhal & Burkard (2005)** showed that only a single global superoperator can describe the entire process, and that the environment retains information about past system states and can feed it back later, thus making errors non-independent.[1]
  - **Aharonov, Kitaev & Preskill (2006)** showed that non-Markovian noise acts jointly across qubits. They reformulate threshold theorems in terms of operator norms rather than independent error probabilities [2].
  - **Lemberger & Yavuz (2017)** showed that for a common bosonic bath, the error per qubit increases with system size, and along with decoherence.[3].
- Pollock *et al.* (2018) speak about a need for a theory of non-Markovian error correction[4].

From Perfect QEC to Approximate QEC

Some Definitions:

The codespace $C := \mathrm{Span}\{|i_L\rangle\}_{i=1}^{2^k}$, the projector $P := \sum_{i=1}^{2^k} |i_L\rangle\langle i_L|$ onto C.

Perfect QEC

- **Knill-Laflamme (KL)[5] conditions:**
  $$PE_i^\dagger E_j P = \alpha_{ij} P$$
- Error subspaces are **orthogonal**:
  $$E_i\mathcal{C} \perp E_j\mathcal{C}, \quad i \neq j$$
- Enables **syndrome measurement**:
  $\mathcal{R}(\rho) = \sum_i U_i^\dagger P_i \rho P_i U_i$,
  $P_i :=$ projector onto $\mathcal{H}_i := E_i C$,
  $\sqrt{\alpha_{ii}} U_i P = E_i P$
- Requires: $n \geq 5$
  (for arbitrary single-qubit noise)

Approximate / Noise-Adapted QEC

- **AQEC condition:**
  $$PE_i^\dagger E_j P = \lambda_{ij} P + PB_{ij} P$$
- Error subspaces are **not orthogonal**
- Syndrome measurements become ambiguous, even worse: **destroy coherence**.
- Recovery could be **CPTP**, but only approximate

Noise-adapting an AQEC Model

- **Fidelity of a Recovery Channel**: Assume a recovery channel $\mathcal{R}$ for an error channel $\mathcal{E}$, we measure worst-case Fidelity

$$F^2_{min}\left[\left|\psi\right\rangle, (\mathcal{R}\circ\mathcal{E})(\left|\psi\right\rangle\left\langle\psi\right|)\right] = \min_{\left|\psi\right\rangle\in C}\left\langle\psi\right|\left((\mathcal{R}\circ\mathcal{E})(\left|\psi\right\rangle\left\langle\psi\right|)\right)\left|\psi\right\rangle. \tag{9}$$

- Given noise $\mathcal{E}\sim\{E_i\}_{i=1}^N$, the optimization problem becomes

$$\max_C \max_{\mathcal{R}} \min_{\left|\psi\right\rangle\in C}\left\langle\psi\right|\left((\mathcal{R}\circ\mathcal{E})(\left|\psi\right\rangle\left\langle\psi\right|)\right)\left|\psi\right\rangle$$

- Mandayam & Ng [6] (2010) showed that the Petz recovery channel, made up of

$$R_i := PE_i^\dagger\mathcal{E}(P)^{-1/2}, \quad P: \text{the projector (unbiased reference state)},$$

achieves perfect QEC if the error follows the KL conditions. Otherwise, it achieves a worst-case fidelity close to that of the optimal recovery channel.

HPTP Noise Model

- Shabani & Lidar [7] (2009) proved by a simple linear argument that any HPTP map $\mathcal{E}(\rho)$ can be written as

$$\mathcal{E}_1(\rho) - \mathcal{E}_2(\rho) = \sum_i A_i \rho A_i^\dagger - \sum_j B_j \rho B_j^\dagger = \sum_\mu \mathrm{sgn}(\mu) K_\mu \rho K_\mu^\dagger, \tag{10}$$

  where both $\mathcal{E}_1(\rho), \mathcal{E}_2(\rho)$ are CP maps.
- Let Δ_{ij} be traceless operators such that

$$\underbrace{P E_i^\dagger(t)(\mathcal{E}(t)[P])^{-1/2}}_{R_i} E_j(t) P = \beta_{ij}(t) P + \Delta_{ij}(t) \tag{11}$$

- Where $\{R_i\}$ are the SK operators for the recovery map $\mathcal{R}_{NM}$.

Bounds on Fidelity

- The recovered state will be given by

$$\sigma := \mathcal{R}_{NM} \circ \mathcal{E}\left[\rho\right] = \sum_{i,j} \mathrm{sgn}(i)\,\mathrm{sgn}(j) R_i E_j P \rho P E_j^\dagger R_i^\dagger, \tag{12}$$

where dependence of R and E on t is implicit, and P is itself $|0_L\rangle\langle 0_L| + |1_L\rangle\langle 1_L|$ (the projector onto the codespace).

- Then, the fidelity loss achieved using non-Markovian Petz recovery is given by

$$\eta_{NM}(t) := 1 - \langle\psi|\sigma|\psi\rangle = \sum_{i,j} \mathrm{sgn}(i)\,\mathrm{sgn}(j) \left[\langle\psi|\Delta_{ij}^\dagger(t)\Delta_{ij}(t)|\psi\rangle - |\langle\psi|\Delta_{ij}(t)|\psi\rangle|^2\right] \tag{13}$$

- We can notice that from the Cauchy-Schwarz inequality, the quantity inside the square brackets is positive. However, the signums attached to them can cause the fidelity loss to fluctuate, which it does.

Bounds on Fidelity (Contd.)

- Now the question arrives, how would one implement such a map. Here, assuming $\mathcal{E}(P)^{-1/2}$ is well-defined (which it is, being unital), we can safely conclude that the channel is HPTP. The real difficulty is in knowing the $E_i(t)$ *at the same time instant* t.
- Due to the practical impossibility in making such a recovery channel, we also consider a simpler, fixed R_i (chosen at a random time t_0), creating the channel $\mathcal{R}_M$. This channel is Markovian and thus is CPTP, so $\mathrm{sgn}(i) = +1$.

$$\eta_M(t) = \sum_{i,j} \mathrm{sgn}(j) \left[\langle\psi|\Delta_{ij}^{\dagger}(t)\Delta_{ij}(t)|\psi\rangle - |\langle\psi|\Delta_{ij}(t)|\psi\rangle|^2 \right] \tag{14}$$

- Through methods like process tomography, we can figure out the SK operators of the $\mathcal{E}$ channel, and thus create the channel $\mathcal{R}_M$ physically.

Simulations: Amplitude Damping Channel

- Considering the Non-Markovian AD channel again, we conduct a simulation of the [[5,1,3]] code[1], with the stabilizer generators

$$S = \langle XZZXI, IXZZX, XIXZZ, ZXIXZ \rangle$$

- To refresh our memory, here are the SK operators for the AD channel:

$$E_1(t) = \begin{pmatrix} 1 & 0 \\ 0 & \sqrt{1-\gamma(t)} \end{pmatrix} \quad ; \quad E_2(t) = \begin{pmatrix} 0 & \sqrt{\gamma(t)} \\ 0 & 0 \end{pmatrix}$$

with damping parameter:

$$\gamma(t) = 1 - |G(t)|^2, \quad G(t) = e^{-bt/2}\left(\tfrac{b}{d}\sinh\left(\tfrac{dt}{2}\right) + \cosh\left(\tfrac{dt}{2}\right)\right), \quad d = \sqrt{b^2 - 2\Gamma_0 b}$$

[1] The astute among the audience may recognize this code from Quiz 2.

Results: Amplitude Damping Channel

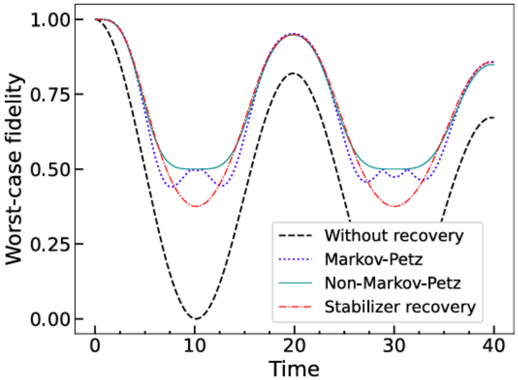


Figure: Performance of different QEC schemes for non-Markovian AD noise with noise parameters $b = 0.01$ and $\Gamma_0 = 5$. We compare the bare qubit fidelity with that achieved by the [[5,1,3]] code with stabilizer recovery and the four qubit code with Petz recovery (Markovian and non-Markovian).

Simulations: Amplitude Damping Channel

- We now consider the 4-qubit code tailored for amplitude damping errors, with codewords

$$|0_L\rangle = \frac{1}{\sqrt{2}}(|0000\rangle + |1111\rangle)$$
$$|1_L\rangle = \frac{1}{\sqrt{2}}(|0011\rangle + |1100\rangle),$$

  and the stabilizer generators

$$S = \langle XXXX, IIZZ, ZZII\rangle$$

- However, the code does not admit a standard stabilizer recovery method, since measuring the stabilizer generators is not sufficient to identify the single-qubit amplitude damping errors unambiguously. Thus, for comparison, we study it with another Noise-adapted map: the Leung recovery map.

Results: Amplitude Damping Channel

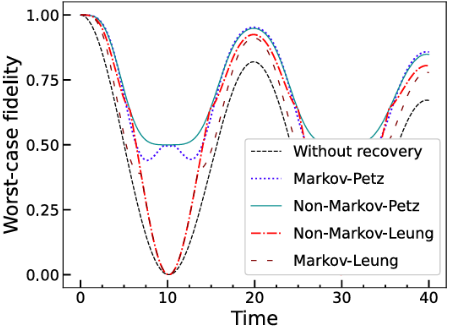


Figure: Performance of different recovery schemes for the [[4,1]] code subject to non-Markovian AD noise with the noise parameters $b = 0.01$ and $\Gamma_0 = 5$. For the Markovian Petz and Markovian Leung recovery, the recoveries are adapted to AD noise in the Markovian regime with $b = 0.1$ and $\Gamma_0 = 0.005$.

The Dephasing Channel

- Here, we consider a non-Markovian dephasing model. The SK operators of this quantum channel are

$$D_1(t) = \sqrt{\frac{1+p(t)}{2}} I \quad ; \quad D_2(t) = \sqrt{\frac{1-p(t)}{2}} Z$$

  with damping parameter:

$$p(t) = e^{-\nu}(\cos(\mu\nu) + \frac{\sin(\mu\nu)}{\mu}), \quad \mu(t) = \sqrt{(4\kappa\tau)^2 - 1}, \quad \nu = \frac{t}{2\tau}$$

- For the codespace, we consider two rep codes of distance 3 and 5,

$$\begin{aligned} |0_L\rangle &= |+++\rangle\,, \quad |1_L\rangle = |---\rangle\,\& \\ |0_L\rangle &= |+++++\rangle\,, \quad |1_L\rangle = |-----\rangle \end{aligned}$$

  with stabs $\langle IZZ, ZZI\rangle$ & $\langle IZZZZ, ZZZIZ, ZZIZZ, ZIZZZ\rangle$ respectively.

Results: Dephasing Channel

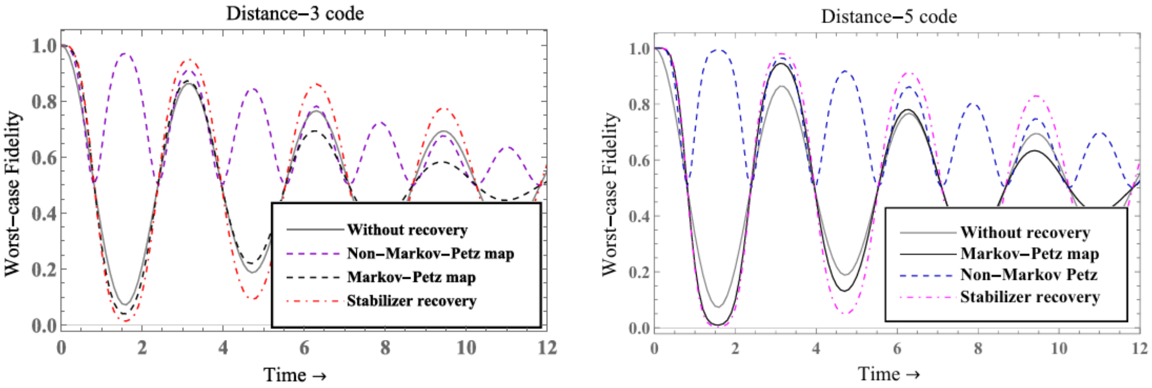


Figure: Performance of different recovery operations with the three-qubit distance-three and five-qubit distance-five code for non-Markovian dephasing noise.

Conclusion

Covered

- Complete description of Non-Markovianity.
- Non-Markovianity as a barrier to QEC.
- Application to Damping and Dephasing models.

Future directions:

- Relationship between eigenvalue spectrum and effectiveness.
- Syndrome-based recovery? "Universal syndrome-based recovery for noise-adapted quantum error correction" - Answered by Same Authors.
- Multiple error-correction cycles: temporal correlations obtained, how do they affect the errors?

Thank You

Questions?

Appendix B: Amplitude damping derivation

- System + bath (Jaynes–Cummings, RWA):

$$H = \omega_0 \sigma_+ \sigma_- + \sum_k \omega_k b_k^\dagger b_k + \sum_k (g_k \sigma_+ b_k + g_k^* \sigma_- b_k^\dagger)$$

- Interaction picture:

$$H_I(t) = \sum_k \left(g_k \sigma_+ b_k e^{i(\omega_0 - \omega_k)t} + g_k^* \sigma_- b_k^\dagger e^{-i(\omega_0 - \omega_k)t} \right)$$

- Single-excitation ansatz:

$$|\Psi(t)\rangle = c_e(t)|e, 0\rangle + \sum_k c_k(t)|g, 1_k\rangle$$

- Leads to memory equation:

$$\dot{c}_e(t) = -\int_0^t d\tau\, f(t-\tau)\, c_e(\tau)$$

Appendix A: Amplitude damping derivation

- Bath correlation:

$$f(t) = \sum_k |g_k|^2 e^{i(\omega_0 - \omega_k)t} \quad \rightarrow \quad \int d\omega \, J(\omega) e^{i(\omega_0 - \omega)t}$$

- Lorentzian spectral density:

$$J(\omega) = \frac{\Gamma_0 b^2}{2\pi[(\omega - \omega_c)^2 + b^2]}$$

- Fourier transform:

$$f(t) = \frac{\Gamma_0 b}{2} e^{-(b - i\Delta)t}, \quad \Delta = \omega_0 - \omega_c$$

- Gives ODE:

$$\ddot{G}(t) + (b - i\Delta)\dot{G}(t) + \frac{\Gamma_0 b}{2} G(t) = 0$$

Appendix A: Amplitude damping derivation

- Solution:

$$G(t) = e^{-(b-i\Delta)t/2}\left[\cosh\left(\frac{dt}{2}\right) + \frac{b-i\Delta}{d}\sinh\left(\frac{dt}{2}\right)\right]$$

$$d = \sqrt{(b-i\Delta)^2 - 2\Gamma_0 b}$$

- Reduced dynamics:

$$\gamma(t) = 1 - |G(t)|^2$$

- Kraus operators:

$$E_1 = \begin{pmatrix} 1 & 0 \\ 0 & \sqrt{1-\gamma(t)} \end{pmatrix}, \quad E_2 = \begin{pmatrix} 0 & \sqrt{\gamma(t)} \\ 0 & 0 \end{pmatrix}$$

- Regimes:

$$b \gg \Gamma_0 \Rightarrow \text{Markovian}, \quad b \lesssim \Gamma_0 \Rightarrow \text{Non-Markovian}$$

Appendix B: Fidelity bound

- HPTP noise:

$$\mathcal{E}(\rho) = \sum_j \mathrm{sgn}(j)\, E_j \rho E_j^\dagger$$

- Petz recovery:

$$R_i = P E_i^\dagger\, \mathcal{E}(P)^{-1/2}$$

- Define AQEC decomposition:

$$R_i E_j P = \beta_{ij} P + \Delta_{ij}$$

  - β_{ij}: ideal (Knill–Laflamme part)
  - Δ_{ij}: traceless deviation

- Input state:

$$\rho = |\psi\rangle\langle\psi|, \quad P|\psi\rangle = |\psi\rangle$$

Appendix B: Fidelity bound

Recovered state:

$$\sigma = \mathcal{R} \circ \mathcal{E}(\rho) = \sum_{i,j} \mathrm{sgn}(i)\mathrm{sgn}(j)\, R_i E_j \rho E_j^\dagger R_i^\dagger$$

Insert projector:

$$\sigma = \sum_{i,j} \mathrm{sgn}(i)\mathrm{sgn}(j)\, (R_i E_j P)\, \rho\, (P E_j^\dagger R_i^\dagger)$$

Using:

$$R_i E_j P = \beta_{ij} P + \Delta_{ij}$$

we get:

$$\sigma = \sum_{i,j} \mathrm{sgn}(i)\mathrm{sgn}(j)\, (\beta_{ij} P + \Delta_{ij}) \rho (\beta_{ij}^* P + \Delta_{ij}^\dagger)$$

Appendix B: Fidelity bound

Fidelity:

$$F^2 = \langle\psi|\sigma|\psi\rangle$$

Since $P|\psi\rangle = |\psi\rangle$:

$$F^2 = \sum_{i,j} \mathrm{sgn}(i)\mathrm{sgn}(j) \left(|\beta_{ij}|^2 + 2\Re[\beta_{ij}^* \langle\psi|\Delta_{ij}|\psi\rangle] + |\langle\psi|\Delta_{ij}|\psi\rangle|^2\right)$$

Using completeness of recovery:

$$\sum_{i,j} \mathrm{sgn}(i)\mathrm{sgn}(j) \left(|\beta_{ij}|^2 + 2\Re[\beta_{ij}^* \langle\psi|\Delta_{ij}|\psi\rangle] + \langle\psi|\Delta_{ij}^\dagger \Delta_{ij}|\psi\rangle\right) = 1$$

Subtracting:

$$F^2 = 1 - \sum_{i,j} \mathrm{sgn}(i)\mathrm{sgn}(j) \left(\langle\psi|\Delta_{ij}^\dagger \Delta_{ij}|\psi\rangle - |\langle\psi|\Delta_{ij}|\psi\rangle|^2\right)$$

$$\boxed{\eta_{NM}(t) = 1 - F^2 = \sum_{i,j} \mathrm{sgn}(i)\mathrm{sgn}(j) \left[\langle\psi|\Delta_{ij}^\dagger \Delta_{ij}|\psi\rangle - |\langle\psi|\Delta_{ij}|\psi\rangle|^2\right]}$$

Appendix C: Some self-simulations

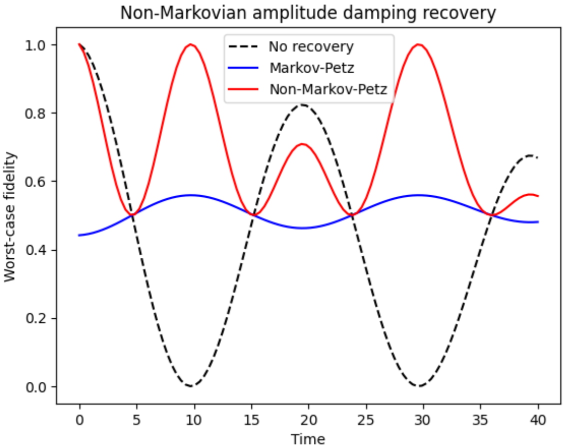


Figure: Simulations on $b = 0.01$, $\Gamma_0 = 5$, $t_0 = 5$

Appendix C: Some self-simulations

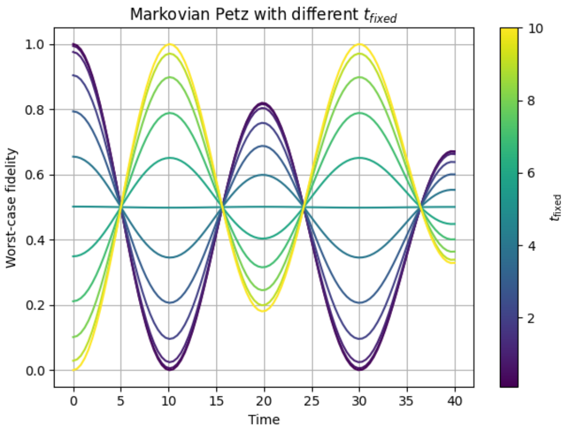


Figure: Simulations of dynamics for $\mathcal{R}_M$ on different t_0s

Appendix C: Some self-simulations

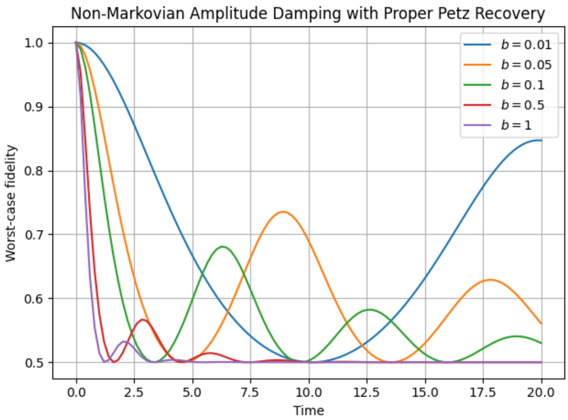


Figure: Effect of b (noise colour) on fidelity

References I

B. M. Terhal and G. Burkard.
Fault-tolerant quantum computation for local non-markovian noise.
Phys. Rev. A, 71:012336, 2005.

D. Aharonov, A. Kitaev, and J. Preskill.
Fault-tolerant quantum computation with long-range correlated noise.
Phys. Rev. Lett., 96:050504, 2006.

B. Lemberger and D. D. Yavuz.
Effect of correlated decay on fault-tolerant quantum computation.
Phys. Rev. A, 96:062337, 2017.

Felix A. Pollock, César Rodríguez-Rosario, Thomas Frauenheim, Mauro Paternostro, and Kavan Modi.
Non-markovian quantum processes: Complete framework and efficient characterization.
Physical Review A, 97(1):012127, 2018.

References II

Emanuel Knill and Raymond Laflamme.
A theory of quantum error-correcting codes.
Physical Review A, 55(2):900–911, 1997.

Hui Khoon Ng and Prabha Mandayam.
Simple approach to approximate quantum error correction based on the transpose channel.
Physical Review A, 81(6):062342, 2010.

Alireza Shabani and Daniel A. Lidar.
Maps for general open quantum systems and a theory of linear quantum error correction.
Physical Review A, 80(1):012309, 2009.